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Deep reinforcement learning enhanced PID control for hydraulic servo systems in injection molding machines

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To address the issue of insufficient position control accuracy in the servo-hydraulic system of injection molding machines under nonlinear characteristics and external disturbances, this paper proposes a novel adaptive PID control strategy enhanced by the Deep Deterministic Policy Gradient (DDPG) algorithm. An auxiliary servo valve is introduced to improve flow capacity and enhance the system's dynamic response performance. Meanwhile, the DDPG algorithm is utilized to adjust the PID parameters in real time based on tracking errors and system state feedback, thereby improving the controller's adaptability to time-varying operating conditions. Compared with traditional control methods, the proposed strategy demonstrates superior tracking accuracy, faster convergence, and stronger robustness. In particular, this work innovatively integrates the DDPG algorithm with an auxiliary servo valve structure for PID parameter optimization and dynamic performance enhancement, offering new ideas and technical pathways for adaptive control of complex hydraulic systems.

Keywords DDPG, Position control, Reinforcement learning, Servo hydraulic system, Intelligent control, Adaptive control

With the growing demand for large injection molded products in aerospace, automotive, defense, and medical industries, the development of efficient, energy-saving, and intelligent large-scale injection molding equipment and associated manufacturing technologies is accelerating. Among the various technical challenges, improving the positioning accuracy of hydraulic cylinders has become particularly critical[1]. As the core component in the molding process, the clamping servo-hydraulic system of an injection molding machine directly affects product quality and production efficiency, making it a key determinant of overall equipment performance[2–5]. Therefore, achieving high-precision, fast-response, and robust control of hydraulic cylinders under dynamic loads and complex environments has emerged as a pressing challenge in the field of servo-hydraulic control. Addressing this challenge is of significant practical importance for advancing the transformation and upgrading of the plastics manufacturing industry[6,7].

PID control methods are widely adopted in servo-hydraulic control systems due to their simple structure and ease of implementation[8–11]. However, PID control relies on manual parameter tuning and struggles to effectively handle system nonlinearities and external disturbances, limiting its optimal performance. This highlights the urgent need for advanced intelligent control strategies to overcome the limitations of traditional methods and enhance the dynamic performance and robustness of servo-hydraulic systems. In recent years, with the continuous development of intelligent control approaches, a variety of effective strategies for nonlinear system control have emerged[12–15]. For instance, Xu et al. proposed an extended-state-observer-based adaptive prescribed performance control method for robust regulation of hydraulic systems under full-state constraints[16]; Wang and Chen developed a model-free optimal containment control method based on the actor-critic framework[17]; Mercorelli and colleagues introduced various optimized observers and robust control strategies for sensorless control of electromagnetic valve actuators[18,19] and servo-hydraulic actuators[20]. In addition, traditional control improvements include ADP-based prescribed-time control strategies[21], PID-fuzzy switching strategies[22], and

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non-lifted norm optimal iterative learning control approaches for networked systems[23], several excellent fixed-structure controller design methods have also been proposed[27–29], providing a solid theoretical foundation for complex dynamic system control. Furthermore, a fault-tolerant control scheme integrating nonlinear disturbance observers and terminal sliding mode control was proposed to enhance the robustness of electro-hydraulic actuators under internal leakage faults and disturbances, demonstrating improved tracking accuracy and system stability[30]. However, fuzzy PID controllers often require expert knowledge to construct rule bases and membership functions, making their design process complex and time-consuming.

Reinforcement Learning (RL), a prominent machine learning approach, has found extensive applications in intelligent control tasks within complex environments[31–38]. In RL, an intelligent agent learns to maximize cumulative rewards by taking actions aimed at achieving a specific objective through interactions with the environment. Deep Reinforcement Learning (DRL), a subfield of RL, leverages high-performance computational resources to offer powerful capabilities in data processing, feature extraction, and generalization, making it a subject of considerable interest among researchers.In 2016, Lillicrap et al. proposed the Deep Deterministic Policy Gradient (DDPG) algorithm[39], which treats nonlinear systems as black-box models without requiring knowledge of their mathematical models or dynamic characteristics. The agent only needs to set an initial and target state, and it can learn an optimal control strategy by interacting with the environment and maximizing the reward function. Compared with traditional methods, DDPG avoids system linearization and complex parameter tuning, which can otherwise degrade performance. As a reinforcement learning algorithm that integrates deep neural networks with deterministic policy gradients, DDPG is well-suited for continuous action spaces and demonstrates excellent learning efficiency, robustness, and generalization ability—making it an effective tool for controlling nonlinear hydraulic systems.

Studies have demonstrated the effectiveness of reinforcement learning-based control methods in optimizing the performance of servo-hydraulic systems. For instance, Tae-Ho Oh et al. proposed a deep reinforcement learning approach based on the Deep Deterministic Policy Gradient (DDPG) algorithm for the simultaneous design and tuning of multiple trap filter[40]. Egli et al. utilized a reinforcement learning algorithm to control the end-effector of a nonlinear hydraulic excavator, achieving superior trajectory tracking accuracy compared to conventional PID control methods[41]. Yao et al. employed an actor neural network to reduce the high feedback gain required by the recursive RISE control method by compensating for unknown dynamics, while integrating a critic neural network to enhance control performance by evaluating the filtered tracking error[24]. In the context of the injection molding machine clamping servo-hydraulic system, where position control precision is critical under varying operational conditions, the development of a control strategy with strong adaptability and robust disturbance rejection capabilities has become essential to enhancing position control accuracy[25–41]. The ability to quickly and accurately adjust PID parameters in response to different working conditions is a key challenge for optimizing system performance in such applications.

Despite recent progress in applying reinforcement learning to improve the control performance of servo-hydraulic systems, several key challenges remain in practical applications.

First, there is a lack of structural adaptability in the hydraulic system design. Most existing studies focus primarily on control algorithms while neglecting the co-optimization of the system structure itself. This issue is particularly evident in asymmetric cylinder scenarios, where flow dynamics play a critical role, yet the design of flow regulation mechanisms is often insufficient. Second, the interpretability and real-time applicability of control strategies remain limited. Although deep reinforcement learning offers model-free and self-learning capabilities, many current approaches directly output control signals without integration with traditional PID structures, which hinders efficient deployment and rapid response in industrial settings. Third, reward function designs are often oversimplified. Many studies use instantaneous error as the sole optimization objective, without adequately considering dynamic performance indicators such as error rate of change or control smoothness. This can result in slow convergence and poor control stability. To address these issues, this study proposes an adaptive control strategy that integrates the Deep Deterministic Policy Gradient (DDPG) algorithm with a traditional PID control structure, while also incorporating an auxiliary servo valve to enhance flow regulation and dynamic response performance. Compared with conventional control methods, the proposed approach retains the simplicity and robustness of PID controllers while leveraging the adaptive learning capabilities of DDPG to achieve real-time optimization under nonlinear and disturbance-prone conditions. Experimental results demonstrate superior performance in terms of control accuracy, convergence speed, and steady-state precision compared with traditional intelligent control methods such as fuzzy control and neural networks. This work thus provides a new perspective for high-precision and highly adaptive control of nonlinear hydraulic systems.

To the best of our knowledge, this is the first study to apply deep reinforcement learning for control optimization in a hydraulic system that integrates both a main control valve and an auxiliary servo valve. This framework provides a new direction for intelligent control in asymmetric hydraulic systems, offering improved adaptability and performance. The main contributions of this paper are summarized as follows:

(1) A composite hydraulic system model integrating an auxiliary servo valve is constructed, which significantly enhances flow regulation and dynamic response performance under asymmetric cylinder scenarios;
(2) An adaptive PID control strategy based on the Deep Deterministic Policy Gradient (DDPG) algorithm is proposed, enabling online optimization of PID parameters and improving the controller's adaptability under complex nonlinear and external disturbance conditions;
(3) A compound reward function is designed by incorporating both error magnitude and error rate of change, which accelerates the convergence of the DDPG policy and improves the stability and efficiency of the learned control strategy.

Problem formulation and system modeling

System description

The system block diagram is shown in Fig. 1, where the controller generates control signals based on displacement feedback to drive the servo valves and actuate the hydraulic cylinder.

To simplify system modeling and focus on control strategy development, the following reasonable assumptions are made during the modeling process:

Incompressible hydraulic oil

The working fluid is assumed to be ideally incompressible, and the effects of fluid compressibility and temperature variation on system performance are neglected (Although the hydraulic oil is assumed to be nearly incompressible, the effective bulk modulus β_e is retained in the dynamic model to account for pressure variation effects. This treatment is consistent with common modeling practices in electro-hydraulic systems.).

Sufficient pipeline rigidity

The hydraulic pipelines are considered to have negligible elastic deformation, and pipeline leakage or pressure losses are not modeled.

Linearized servo valve dynamics

The responses of both the main and auxiliary valves are approximated as first-order linear systems, and only their static gain characteristics are considered in the model.

Neglect of sensor delays

Displacement and pressure sensors are assumed to operate without time delay, and the sampling frequency is sufficient to meet the dynamic response requirements of the system.

To enhance the system's flow regulation capability and dynamic response under high-performance demands, and to suppress the startup delay caused by the structural asymmetry of the hydraulic cylinder, an auxiliary servo valve is introduced to form a dual-valve control hydraulic circuit. This design draws inspiration from dual-valve configurations commonly used for asymmetric cylinders, in which the auxiliary valve is arranged in parallel with the main control valve to compensate for the initial flow imbalance between the rod and rodless chambers. This structure effectively mitigates startup delays and improves the system' dynamic response.

The specific control logic is as follows: when the piston rod extends, the auxiliary valve assists the main valve in supplying oil to the rodless chamber; when the piston retracts, the auxiliary valve remains inactive. This configuration not only enhances the flexibility of oil supply but also optimizes the flow distribution between the rod and rodless chambers during the initial phase, thereby improving the overall dynamic performance of the hydraulic system.

Servo valve flow equations

The dual-valve position servo hydraulic system considered in this study is shown in Fig. 1, and the dynamics of the inertial load are given in Eq. (1):

$$M\,\ddot{x}_p = P_L A - B\dot{x}_p - f\left(x_p, \dot{x}_p\right) - d_D\left(t\right) \tag{1}$$

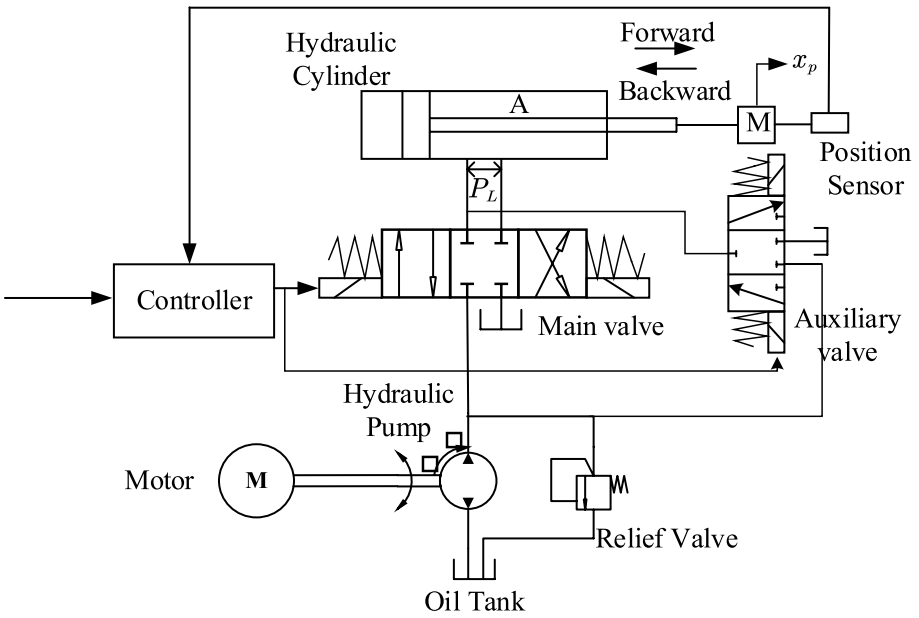


Fig. 1. Schematic diagram of a dual-valve position servo hydraulic system.

In this context, M and x_p denote the mass and position of the load, respectively. P_L represents the pressure differential of the hydraulic cylinder, A indicates the effective area of the hydraulic cylinder, and B signifies the viscosity coefficient. The term $f\left(x_p, \dot{x}_p\right)$ characterizes the nonlinear friction of the unmodeled dynamics, where $d_D\left(t\right)$ represents the non-matching disturbance.

The rate of change of hydraulic pressure $\dot{P}_L$ is defined as follows[13,16,39].

$$\frac{V_t}{4\beta_e}\dot{P}_L = -A\dot{x}_p - C_t P_L + q\left(t\right) + Q_L \tag{2}$$

In which V_t represents the volume of the hydraulic system, β_e denotes the compressibility of the fluid, C_t indicates the hydraulic system's flow resistance coefficient, and $q\left(t\right)$ represents the modeling error. The output flow rate Q_L is established as follows.

$$Q_L = k_q x_v \sqrt{P_s - sign\left(x_v\right) P_L} \tag{3}$$

here $k_q = C_d\omega\sqrt{1/\rho}$ represents the flow gain, P_s denotes the hydraulic pressure, and the function $sign\left(x_v\right)$ is defined as.

$$sign\left(x_v\right) = \begin{cases} 1, if\, x_v \geq 0 \\ -1, if\, x_v < 0 \end{cases} \tag{4}$$

Due to the high gain of the actuator, we can effectively neglect its dynamic characteristics and treat x_v as a proportional input to the controller, specifically $x_v = k_i u$(valve approximation) with $k_i > 0$ being the gain constant[39]. Thus, based on (3), we can derive the flow rate of the actuator as:

$$Q_L = k_t u \sqrt{P_s - sign\left(u\right) P_L} \tag{5}$$

where $k_t = k_q k_i$.

Similar to the main valve, the auxiliary valve can be modeled as a proportional orifice flow element, and its output flow rate Q_a is given by:

$$Q_a = k_{ta} \cdot u_a \cdot \sqrt{P_s - sign(u_a) P_L} \tag{6}$$

Although both the main and auxiliary valves are driven by the same control signal $_u$, they operate on different flow paths (rod chamber vs. rodless chamber) and have distinct flow gain coefficients k_t and k_{ta}, allowing for effective compensation of initial flow imbalance in the dual-valve system.

To preserve mathematical tractability and prevent non-differentiability, the function $sign(u)$ can be approximated using a smooth surrogate, such as $\tanh(ku)$ which retains directional information while ensuring continuity.

The total flow into the cylinder is therefore defined as:

$$Q_{total} = Q_L + Q_a \tag{7}$$

Controller design and reinforcement learning strategy

Comparison between DDPG and other reinforcement learning algorithms

To address the continuous control requirements of the servo-hydraulic positioning system in injection molding machines, this study adopts the Deep Deterministic Policy Gradient (DDPG) algorithm as the core of the intelligent control strategy. Compared with discrete-action reinforcement learning algorithms such as Deep Q-Network (DQN), DDPG can directly output real-valued actions, eliminating quantization errors caused by action discretization. This makes it particularly suitable for electro-hydraulic systems characterized by strong nonlinearity, real-time dynamics, and high precision demands. In addition, although Proximal Policy Optimization (PPO) has recently gained attention due to its stability in policy updates, it is fundamentally a stochastic policy method, generating actions with inherent probabilistic noise. For industrial servo-hydraulic systems, control commands must be stable and reproducible, which necessitates deterministic outputs to ensure consistent and safe system responses. In contrast, DDPG utilizes a deterministic policy, where each state maps to a unique control action, making it more suitable for real-world industrial scenarios.

Overall, DDPG offers three key advantages over both DQN and PPO in this context:

(1) *Efficient learning in continuous action domains* DDPG leverages policy gradients and deterministic updates to enable stable and sample-efficient training in environments where actions cannot be discretized meaningfully.
(2) *Improved convergence and robustness* The use of target networks and experience replay buffers in DDPG mitigates policy oscillations and overestimation biases, enhancing convergence speed and control robustness, particularly in nonlinear and disturbance-prone systems.
(3) *Deterministic output aligned with industrial control requirements* Unlike stochastic policies, DDPG provides consistent and predictable control outputs, which is essential for maintaining stability in safety–critical hydraulic systems.

DDPG reinforcement learning algorithm

The DDPG algorithm is built upon the Actor-Critic framework and is designed for continuous state and action spaces. In this approach, two neural networks are employed: an Actor network that outputs continuous actions given a state, and a Critic network that evaluates the quality of actions through Q-values. Let $\mu(s|\theta^{\mu})$ denote the deterministic policy represented by the Actor network with parameters θ^{μ}. Let $Q(s,a|\theta^{Q})$ denote the action-value function, which is approximated by the Critic network parameterized by θ^{Q}. The Critic evaluates the expected return for taking action a in the state s under the current policy. Target networks $Q'(s,a|\theta^{Q'})$ and $\mu'(s|\theta^{\mu'})$ are also maintained to stabilize the training process by slowly tracking the learned networks.

At each time step t, the agent receives a state s_t, selects an action $a_t = \mu(s_t|\theta^{\mu}) + N_t$(with added exploration noise N_t), and receives a reward r_t and the next state s_{t+1}. The experience tuple (s_t, a_t, r_t, s_{t+1}) is stored in the replay buffer. A mini-batch of samples is randomly selected to update the networks.

y_j denotes the target Q-value used to update the Critic network during the j-th training iteration. It is computed as:

$$y_j = r_j + \gamma \cdot Q'(s_{j+1}, \mu'(s_{j+1}|\theta^{\mu'})|\theta^{Q'}) \tag{8}$$

where r_j is the immediate reward at step j, γ is the discount factor, Q' is the target Critic network, and μ' is the target Actor network.

The Critic network is updated by minimizing the loss function:

$$Loss = \frac{1}{N}\sum [y_j - Q(s_j, a_j|\theta^{Q})]^2 \tag{9}$$

where N is the batch size.

The Actor network is trained to maximize the expected return:

$$J = E_{s\sim\rho^{\beta}}[Q(s, \mu(s|\theta^{\mu}))] \tag{10}$$

The policy gradient used to update θ^{μ} is:

$$\nabla_{\theta^{\mu}} J \approx E_{s\sim\rho^{\beta}}[\nabla_a Q(s,a|\theta^{Q})|_{s=s_t, a=\mu(s_t)} \cdot \nabla_{\theta^{\mu}} \mu(s|\theta^{\mu})] \tag{11}$$

here ρ^{β} derives from the experience replay buffer, μ denotes the policy strategy, and θ represents the neural network parameters for both the Actor and Critic.

The Bellman equation used for updating the Critic network is:

$$Q_{\phi}(s_t, a_t) = E_{s_{t+1}\sim p(s_{t+1}|s_t,a_t)}[r(s_t, a_t) + \gamma \cdot Q_{\phi'}(s_{t+1}, \mu'(s_{t+1}))] \tag{12}$$

where $P(s_{t+1}|s_t, a_t)$ denotes the state transition probability and $r(s_t, a_t)$ the reward function, the action a_{t+1} is obtained by evaluating the target policy μ′ at the state s_{t+1}, ensuring smooth target estimation for the Critic's update, and the action $a_{t+1} = \mu(s_{t+1}|\theta^{\mu})$ is selected using the target policy network.

The parameters of the target networks are updated using soft updates:

$$\begin{aligned} \theta^{Q'} &\leftarrow \tau \cdot \theta^{Q} + (1-\tau) \cdot \theta^{Q'} \\ \theta^{\mu'} &\leftarrow \tau \cdot \theta^{\mu} + (1-\tau) \cdot \theta^{\mu'} \end{aligned} \tag{13}$$

where τ∈(0,1) is typically set to a small value (e.g., 0.001) to ensure smooth updates.

Overview of the control strategy

In industrial control of servo-hydraulic systems, conventional PID controllers remain the dominant choice due to their simple structure, low computational requirements, and ease of implementation on PLC based platforms. Their transparent control logic and well understood parameter influence make them highly interpretable and practical in engineering applications. However, traditional PID tuning methods such as Ziegler–Nichols or manual trial and error are often based on linear assumptions and fixed parameters, which fall short when applied to the strongly nonlinear and time-varying nature of hydraulic systems. Moreover, these tuning methods typically rely on accurate modeling or expert experience, making them vulnerable to performance degradation under unpredictable disturbances such as load changes, friction variations, or temperature shifts.

To address these challenges, we retain the classical PID control structure as a reliable and interpretable foundation, while introducing the Deep Deterministic Policy Gradient (DDPG) algorithm to enable online adaptation of PID gains. This hybrid architecture, illustrated in Fig. 2, combines the proven advantages of PID fast response and structural simplicity with the adaptive learning capability of DDPG in high-dimensional, model free environments. By doing so, the controller can maintain stable and precise tracking performance even under complex and uncertain conditions, making it suitable for practical deployment in intelligent hydraulic systems.

In this control strategy, the observation space includes the position error, the error rate, and the velocity of the target displacement, while the action space consists of the three PID control parameters. The upper-layer DDPG controller continuously adjusts the parameters of the lower-layer PID controller based on real-time system feedback, enabling adaptive response to external disturbances and achieving faster and more accurate

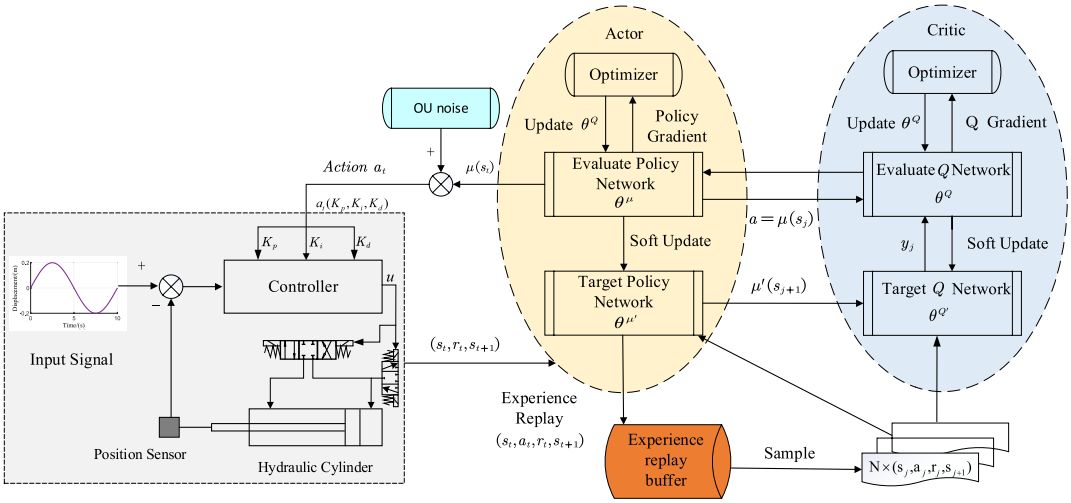


Fig. 2. Main block diagram of the DDPG-PID control strategy.

trajectory tracking. Each hidden layer in the neural network uses a Rectified Linear Unit (ReLU) as the activation function to enhance the mapping capability from input features to control outputs. Figure 2 illustrates the overall structure and feedback mechanism of the DDPG-PID control strategy, and the discretized PID control formulation is presented in Eq. (14).

$$u(k) = K_p e(k) + K_i \sum_{n=0}^{k} e(k) + K_d (e(k) - e(k-1)) \tag{14}$$

Reward function design

In the reinforcement learning method in this paper, the rewards are provided by the servo-hydraulic system environment, which aims to ensure the accurate positioning of the hydraulic cylinder with reduced overshoot. According to the specific requirements of the target task of the intelligent body, the design of the reward function needs to be combined with the task characteristics and the environmental interaction state to ensure that the model performs as expected. The reward function design based only on whether the servo-hydraulic system reaches the target point range in each round is a sparse reward strategy, which may not be able to adequately guide the effective learning of the intelligent body in complex environments. Therefore, this paper proposes a composite reward function to improve the learning efficiency and convergence performance of the model.

$$R = R_1 + R_2 \tag{15}$$

In Eq. (15), R_1 is the reward for the displacement error of the hydraulic cylinder actuator,R_2 is the reward for the slope error of the actuator.

$$R_1 = \begin{cases} -a * \log(\max(eps, e_x)), & if\ e_x < 0.003 \\ -700 * (e_x - 0.003), & if\ e_x > 0.003 \end{cases} \tag{16}$$

$$R_2 = \begin{cases} 0, & if\ e_v < 0.05 \\ -b * (e_v - 0.05), & else \end{cases} \tag{17}$$

where e_x is the displacement error, e_v is slope error, error tolerance, and penalty factor $a \in (0.1, 1), b \in (1, 100)$ can be adjusted according to the actual situation. In this study, the design of the reward function aims to encourage the system to converge quickly. Therefore, we do not want the magnitude of the rewards or penalties to be too large, as it could affect the stability of the agent's learning. Based on empirical adjustments and considering the overall situation, we set a = 0.1 and b = 10 in this paper.

Compared with conventional reward function designs, such as sparse rewards (reward only upon reaching a goal) or purely error-based penalties, the proposed composite reward function provides richer feedback signals to the DDPG agent. Sparse reward schemes tend to result in poor exploration and slow convergence, especially in high-dimensional and nonlinear control tasks. Purely displacement-based penalties, while simple, often fail to encourage dynamic performance improvements such as fast response or minimal overshoot. By incorporating both the magnitude of the tracking error and its slope, our reward function enables more efficient policy learning, helping the agent converge more quickly and achieve higher control accuracy in complex servo-hydraulic environments.

Algorithmic pseudo-code and full workflow

Figure 2 illustrates the working structure of the DDPG-PID control strategy, in which the reinforcement learning algorithm execution process consists of inputting the servo-hydraulic system environment E, the state space S, the action space A, and initialization operations on the neural networks of the Actor and Critic. Subsequently, the loop training process is started. The algorithm pseudo-code is as follows, and the full workflow is shown in Fig. 3:

Initialize the relevant parameters of the policy network (Actor), Q network (Critic), and replay buffer size.
for episode = 1,2,3, …,E **do**
Receive the initial state s_0
for step = 1,2,3 …, T **do**
Compute error: $e(t) = x_f(t) - x_{real}(t)$
Compute control signals: $u(t) = K_p e(t) + K_i \sum_{n=0}^{t} e(t) + K_d \left(e(t) - e(t-1)\right)$
Select actions a_t based on the current state s_t with exploration noise N_t: $a_t = \pi(s_t \mid \theta) + N_t$
Compute the control signals u_t according to the action a_t
Apply control signals u_t and observe the next state s_{t+1}
Compute the current reward r_t
Store following transition (s_t, a_t, r_t, s_{t+1}) into the Experience replay buffer
If it is time to update **then**
Sample mini-batch transitions $(s_j, a_j, r_j, s_{j+1})s$ from the replay buffer.
Update Q-network parameters: $Q_j = Q_j + \lambda_Q \left(r_t + \gamma * Q'(s_{t+1}, a_{t+1}) - Q(s_t - a_t)\right)$
Update Actor network parameters: $\theta \leftarrow \theta + \lambda_\theta \nabla_\theta J(\theta)$
Update target network parameters online.
End if
End for
End for

Algorithm 1. Pseudo-code of the DDPG-PID control strategy.

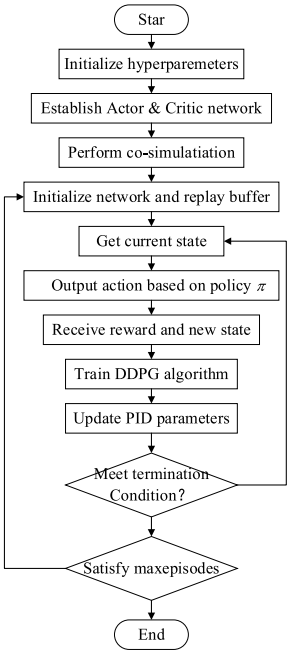


Fig. 3. Full workflow of the DDPG-PID approach.

During each training cycle, the algorithm first receives the initial state and computes the current error. Based on this error, the control signal is calculated using a PID controller, with exploration noise added to generate executable actions. Subsequently, the control signal is applied to the system, and the next state and its corresponding reward are observed. The algorithm records the transition of the current state, action, reward, and next state into the experience replay buffer. When conditions are met, the algorithm samples mini-batch transitions from the replay buffer and updates the parameters of the Critic network and Actor network to continuously optimize the control strategy. This process continues until the predetermined number of training iterations is reached.

Simulation setup and experimental results

Joint simulation model environment

This study employs a co-simulation framework integrating Simcenter AMESim and MATLAB/Simulink, with each platform serving distinct purposes: AMESim is responsible for modeling the servo-hydraulic system, while MATLAB/Simulink is used for designing and training the reinforcement learning controller. AMESim, as a professional modeling tool for hydraulic and mechanical systems, is used to construct a high-fidelity physical model of the valve-controlled asymmetric hydraulic cylinder, including components such as the hydraulic pump, proportional servo valve, and cylinder. The parameter configurations of each subsystem are detailed in Table 1. Compared to traditional analytical modeling approaches, AMESim modeling eliminates the need to derive complex mathematical models or transfer functions of the hydraulic system, significantly reducing modeling complexity.

The reinforcement learning controller is developed using the Reinforcement Learning Toolbox in MATLAB/Simulink. A DDPG agent generates control policies by dynamically tuning the PID parameters in real time. Real-time data exchange between AMESim and Simulink is achieved through a dedicated co-simulation interface, ensuring efficient closed-loop integration of system states, control actions, and feedback responses. The co-simulation runs on a workstation equipped with an Intel i5-13400F CPU and 32 GB RAM, providing sufficient computational resources and simulation efficiency.

As illustrated in the Fig. 4, the joint simulation model utilizes the displacement and velocity signals of the hydraulic cylinder's piston rod as input ports, while the control signals of the servo valve serve as output ports. This design enables real-time interaction between the combined-mode servo-hydraulic system and MATLAB/Simulink through the development of a joint simulation interface module in AMESim.

The combined-mode servo-hydraulic system consists of a motor, a fixed-displacement pump, and a relief valve serving as a safety valve for the constant pressure oil supply unit. The servo directional valve acts as the control element, while the single-rod double-acting hydraulic cylinder serves as the actuator. Displacement and velocity sensors provide real-time feedback on the changes in displacement and speed of the hydraulic cylinder. The feedback displacement change is then compared with the desired displacement change of the hydraulic cylinder to determine the error. Through joint simulation in MATLAB/Simulink, the control algorithm is implemented to achieve the opening control of the directional valve, thereby regulating the hydraulic oil flow that moves the piston rod within the hydraulic cylinder, enabling position control of the hydraulic cylinder's piston rod. The joint simulation model constructed under MATLAB/Simulink is shown in Fig. 5.

To more comprehensively demonstrate the control system's performance, a heuristic analysis of the stability of the DDPG-PID control strategy is provided. The control structure relies on the reinforcement learning agent to adjust the PID parameters online. After convergence during training, the actions output by the Actor remain within a stable range. Combined with a stable Simulink simulation model and the dynamic response of the AMESim hydraulic system, the overall system did not exhibit divergence or oscillatory instability across multiple testing scenarios, indicating that the strategy maintains good dynamic stability before real-world deployment.

Name/unit	Value
Motor speed n_1(r/min)	1450
Displacement volume V(cc/rev)	100
Pump speed n_2(r/min)	1800
Relief valve opening pressure P/bar	300
Main valve frequency w_{sv}/Hz	80
Auxiliary valve frequency w_{sa}/Hz	80
Active cylinder diameter D/mm	120
Rod diameter d/mm	60
Total hydraulic line length L/mm	500
Load mass M/kg	100
Fluid damping coefficient H_c/[L/min/bar]	7×10^{-11}
Viscosity resistance coefficient B_c/[N/(m/s)]	400
Liquid bulk modulus Pa	7×10^{-8}

Table 1. Main parameters of the servo hydraulic system.

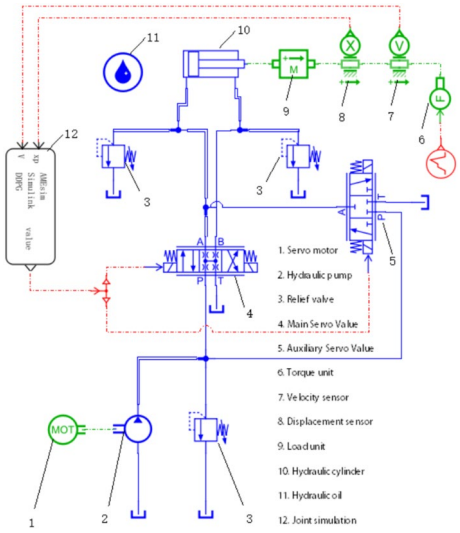


Fig. 4. AMESim co-simulation model of the servo hydraulic system.

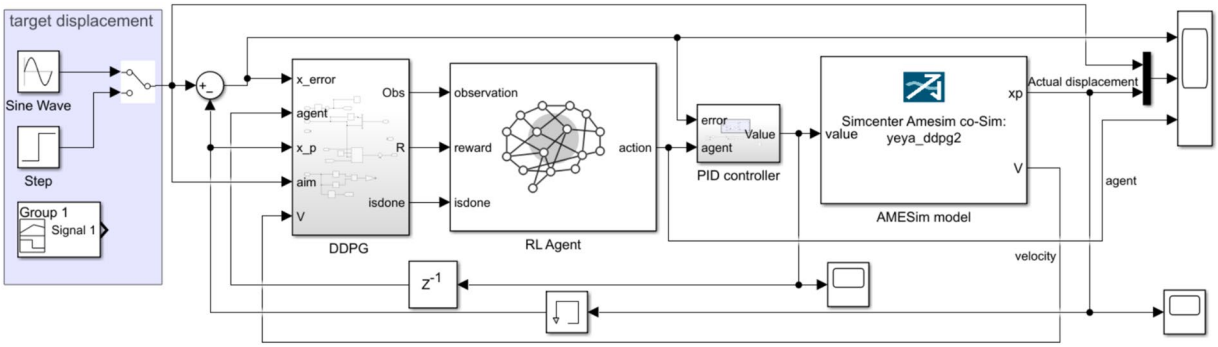
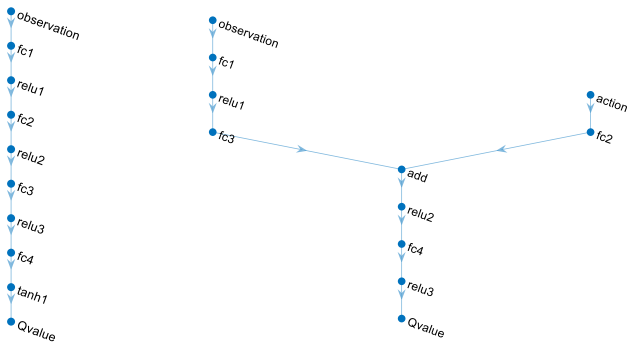


Fig. 5. Simulink model of the DDPG-PID control strategy.



(a) (b)

Fig. 6. The neural network structure layer defined by DDPG.

The establishment of the network and algorithm parameters

In the DDPG control framework proposed in this study, both the Actor and Critic are implemented as multi-layer fully connected neural networks, as illustrated in Fig. 6a shows the structure of the Actor network, where the input consists of the observed system states (i.e., position error, error rate, and target velocity). These inputs are processed through four fully connected layers (fc1–fc4) with ReLU activation functions for feature extraction. The final output layer uses a tanh activation function to produce normalized real-valued actions,

Hyperparameter	Value
Buffer size	1e6
Mini-batch size	256
Actor LEARNING RATE	0.0003
Critic learning rate	0.0003
Discount factor	0.99
Max training steps	1000
Maximum training rounds	500

Table 2. Hyperparameters of the DDPG algorithm.

Layer	Actor	Critic
Input	3-dimensional state	3 state + 3 action
Hidden layer 1	fc1 (200 neurons), ReLU	fc1 (128 neurons), ReLU
Hidden layer 2	fc2 (200 neurons), ReLU	fc3 (128 neurons), ReLU
Hidden layer 3	fc3 (200 neurons), ReLU	fc4 (128 neurons), ReLU
Hidden layer 4	fc4 (200 neurons), ReLU	fc5 (128 neurons), ReLU
Fusion layer	–	Add(state branch + action branch)
Output	3-dimensional action, Tanh	1-dimensional Q-value, Linear

Table 3. Neural network parameters for actors and critics.

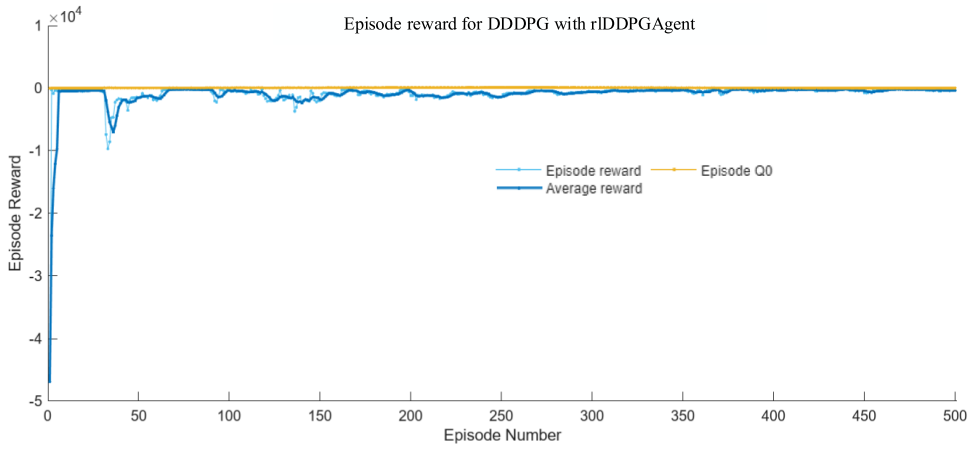


Fig. 7. Training convergence plot of the DDPG algorithm.

corresponding to the three PID control parameters. Figure 6b presents the structure of the Critic network, which adopts a dual-input architecture: the state and action information are processed separately through independent fully connected layers, followed by an addition layer (add) to merge the feature representations. The combined features are further processed through three hidden layers with ReLU activations to finally produce the estimated Q-value.

All hidden layers in the neural networks use ReLU activation functions to mitigate gradient vanishing and enhance nonlinear representation capability. The output layer of the Actor network adopts a tanh activation function to constrain the output range, while the Critic's output layer is linear.

Regarding hyperparameter tuning, considering the strong nonlinearity and sensitivity to control delay in servo-hydraulic systems, the parameters were systematically optimized based on literature recommendations and preliminary experimental evaluations. The final parameter configuration is listed in Table 2, including learning rates, discount factor, soft update coefficient, replay buffer size, and batch size, which collectively balance convergence speed and training stability. The architectural details of the neural networks are provided in Table 3.

Figure 7 shows the convergence curve of the cumulative episode reward during the training process of the DDPG agent. It can be observed that the agent quickly discovered a high-performance initial control policy within the first 10 episodes. Although slight fluctuations occurred in some subsequent episodes, the reward curve remained consistently stable in the second half of the training process without significant variation, indicating that the policy had gradually converged. This curve validates the learning efficiency and convergence

performance of the DDPG algorithm in the continuous action space of the hydraulic system, and further demonstrates that the designed reward function effectively and rapidly guides the policy toward the desired optimization objective.

During the entire simulation process, the displacement signal of the hydraulic cylinder's piston rod is detected and utilized as a feedback signal. The difference between the feedback signal and the desired signal serves as one of the inputs to the control system and is transmitted to the DDPG agent. This agent receives three data components from the DDPG environment: state, reward, and termination condition, allowing it to continuously adjust the output of the three PID control parameters in real time. This adjustment drives the operation of the electro-hydraulic proportional valve, enabling the motion of the piston rod within the hydraulic cylinder. The resulting displacement is fed back into the system, creating a closed-loop process that ultimately fulfills the position control function of the electro-hydraulic proportional system.

Comparison and analysis of simulation results

To validate the performance of the DDPG-PID control strategy for position control in the servo-hydraulic system, three sets of simulation tests are designed[8,13]. First, in the initial simulation, fuzzy PID control and sliding mode control—widely used methods for addressing mechanical system control issues—are selected for comparative analysis to assess the dynamic performance and robustness of the proposed strategy. In the second simulation, the effectiveness of the proposed control strategy in tracking dynamic signals is evaluated using a sine wave as the input. To further verify its practicality in real engineering applications, the simulation tests are designed around the steps of the injection molding process, specifically the closing, locking, and opening phases, with two critical points selected for dynamic performance analysis. All three experimental models are implemented in AMESim and MATLAB/SIMULINK, with a simulation duration of 10 s and a time step of 1 ms.

In this study, the proposed DDPG-PID controller employs a reinforcement learning strategy to optimize the parameters of the proportional-integral-derivative (PID) controller in real time. To prevent high-frequency noise and control oscillations that may arise from the derivative term under limited sensor sampling conditions, the derivative component is omitted in all control methods, and thus K_d is consistently set to zero. To ensure training stability and the physical realizability of the controller output, the action range of the agent is constrained by limiting K_p and K_i within the range of [0, 100].

Experiment 1: response characteristics of the controller

In a precise model, with information derived from an uncontrollable external environment, a comparison is made between the performance of PID control methods and nonlinear sliding mode control methodologies. For the sliding mode controller, the control expression is as follows:

$$value = -a\,(x - x_d) - \varepsilon \cdot sign\,(s) - k \cdot s \tag{18}$$

where the sliding variable is defined as $s = a\,(x - x_d)$, a is the control gain, ε represents the system's position deviation,k is a system parameter,x denotes the actual system position, and x_d indicates the desired position. The parameters used in the sliding mode controller in this paper are configured as follows: $g = 9.81, a = 30, k = 30, \varepsilon = 0.01$.

For fuzzy PID, set the input variables e and ec, and the fuzzy subsets of the output variables $\Delta K_p, \Delta K_i, \Delta K_d$ are all {NB, NM, NS, ZO, PS, PM, PB}. Where the total stroke of the hydraulic cylinder is 0.5 m and the initial position of the rod is 0.25 m, therefore $e \in[-250,250]$ and $ec \in[-250,250]$. The adjustments to the gains are defined as $\Delta K_p \in[-0.3,0.3]$,$\Delta K_i \in[-0.1,0.1]$, and $\Delta K_d \in[-0.0005,0.0005]$. Figure 8 shows the fuzzy control logic, and Table 4 presents the fuzzy control rules.

Figure 9 shows the simulation comparison results of the proposed strategy with fuzzy PID and sliding mode control algorithms on the response of the step displacement signal with 0.2 as the final value. More detailed control performance metrics and evaluation results are listed in Table 5. According to Fig. 9a, the performance of the hydraulic system in terms of step signal position control, the sliding mode control method took 1.13 s to reach the target position, the fuzzy PID control method reached the target displacement in 0.97 s but there was overshooting, and the DDPG-PID control method had a rise time of 0.93 s, with a fast and overshoot-free system response.

To verify the robustness of the proposed method, three different disturbance signals were introduced during the step response process. Figure 9b shows the results of the robustness experiment. Step, ramp, and random-valued disturbance pulses were respectively added at 3 s, 5 s, and 7 s. The disturbance settings are as follows:

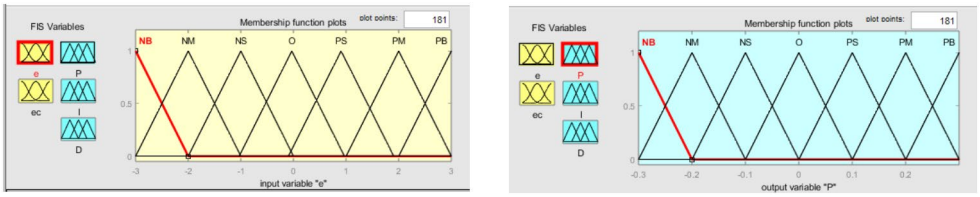


Fig. 8. Fuzzy logic control.

	e																						
	ΔK_p							ΔK_i							ΔK_d								
ec	NB	NM	NS	ZO	PS	PM	PB	NB	NM	NS	ZO	PS	PM	PB	NB	NM	NS	ZO	PS	PM	PB		
NB	PB	PB	PM	PM	PS	ZO	ZO	NB	NB	NM	NM	NS	ZO	ZO	PS	NS	NB	NB	NB	NB	PS		
NM	PB	PB	PM	PS	PS	ZO	NS	NB	NB	NM	NS	NS	ZO	ZO	PS	NS	NB	NM	NM	NM	ZO		
NS	PM	PM	PM	PS	ZO	NS	NS	NB	NM	NS	NS	ZO	PS	PS	ZO	NS	NM	NM	NS	NS	ZO		
ZO	PM	PM	PS	ZO	NS	NM	NM	NM	NM	NS	ZO	PS	PM	PM	ZO	NS	NS	NS	MS	NS	ZO		
PS	PS	PS	ZO	NS	NS	NM	NM	NM	NS	ZO	PS	PS	PM	PB	ZO	ZO	ZO	ZO	ZO	ZO	ZO		
PM	PS	ZO	NS	NM	NM	NM	NB	ZO	ZO	PS	PS	PM	PB	PB	PB	NS	PS	PS	PS	PS	PB		
PB	ZO	ZO	NM	NM	NM	NB	NB	ZO	ZO	PS	PM	PM	PB	PB	PB	PM	PM	PM	PS	PS	PB		

Table 4. $\Delta K_d, \Delta K_d, \Delta K_d$ fuzzy rule control table.

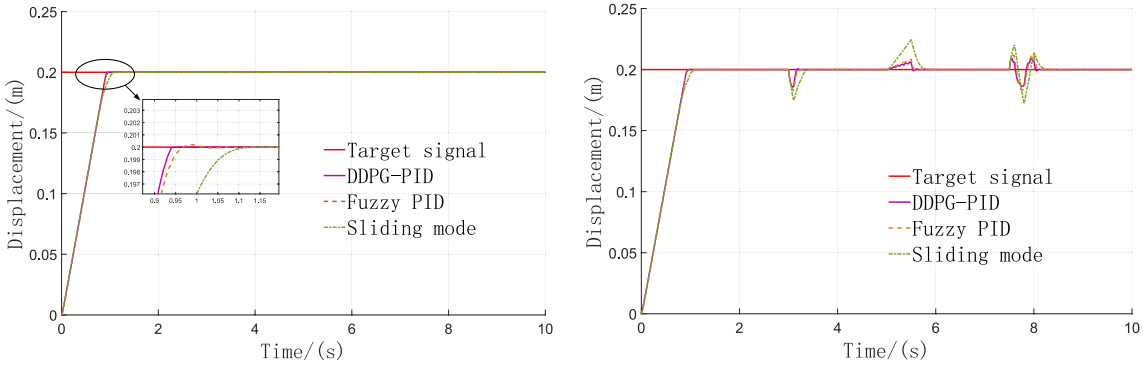


Fig. 9. Hydraulic cylinder step displacement signal response.

	Rising time/s	Adjustment time/s	Overshoot
Sliding mold	1.13	1.13	0.000011
Fuzzy PID	0.97	0.97	0.0003
DDPG-PID	0.93	0.93	0.000016

Table 5. Control performance indicators.

- *Step disturbance* The amplitude is set to 50, simulating a sudden impact equivalent to 50% of the rated load on the system.
- *Ramp disturbance* With an initial value of 0, a final value of 50, and a slope of 1, it represents a gradually varying load condition applied to the system.
- *Random pulse disturbance* Constructed by multiplying a Uniform Random Number within the range (−0.2, 0.2) with a step signal of amplitude 50, this simulates a bounded random disturbance input commonly encountered in real-world systems.

As shown in Table 6, across three distinct disturbance scenarios, the DDPG-PID controller consistently outperforms Fuzzy PID and Sliding Mode in terms of lower peak deviation, faster recovery time, and lower RMSE, demonstrating strong robustness under various nonlinear disturbance conditions.

From an engineering perspective, the reduction in rise time and adjustment time directly benefits the production process of injection molding machines. Faster actuator response translates to shorter clamping and release phases, effectively reducing the cycle time of each injection molding operation. This improvement contributes to increased production efficiency. Moreover, quicker stabilization of the actuator reduces oscillations and mechanical shocks, which helps minimize wear and maintenance frequency, thereby improving the overall reliability and longevity of the system in industrial use.

Experiment 2: sinusoidal signal response simulation

Figure 10a presents the system response under a sinusoidal displacement input with an amplitude of 0.2 m and frequency of 0.63 rad/s. To ensure consistency between the signal and cylinder motion, the initial piston position was set to 0.25 m in AMESim with a total stroke of 0.5 m.

External disturbances	Controller	Peak error (mm)	Recovery time (s)	RMSE (mm)
Step	DDPG-PID	14.252	0.17	0.00963
	Fuzzy PID	17.17	0.26	0.009996
	Sliding Mode	25.07	0.43	0.01261
Ramp	DDPG-PID	6.155	053	0.0035
	Fuzzy PID	8.505	0.61	0.004719
	Sliding Mode	24.463	0.9	0.01176
Random-valued pulses	DDPG-PID	13.779	0.54	0.00829
	Fuzzy PID	16.634	0.65	0.01065
	Sliding Mode	27.491	0.8	0.01193

Table 6. Control Performance Indicators With External Disturbances.

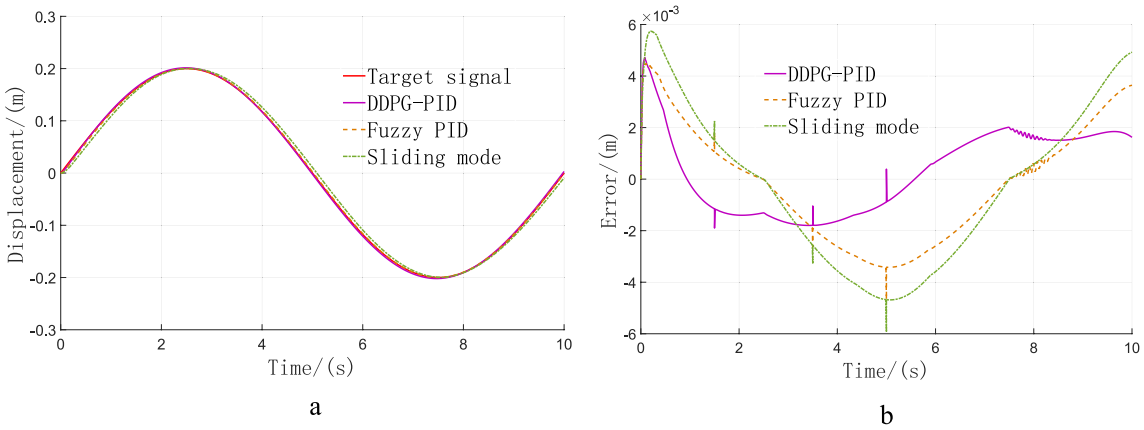


Fig. 10. Hydraulic cylinder sine displacement signal response.

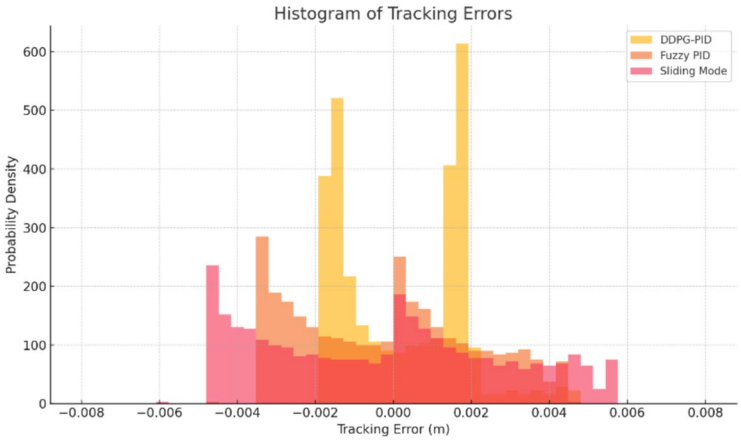


Fig. 11. Tracking error density comparison over 10 s simulation.

Figure 10b shows the tracking error comparison. The maximum error for sliding mode control exceeds 5 mm, and that of fuzzy PID surpasses 3 mm, while the DDPG-PID controller maintains errors within 2 mm—improving accuracy by over 60% and 33.3% compared to the sliding mode and fuzzy PID controllers, respectively. All three curves show small transient spikes due to rapid control adjustments and unfiltered sensor signals, but their amplitudes are negligible.

Overall, the DDPG-PID strategy achieves the highest tracking precision with minimal error, confirming its effectiveness in sinusoidal position control.

Figure 11 presents the histogram of tracking errors in the hydraulic servo system during a sinusoidal signal tracking task. The horizontal axis represents the position error (in meters), while the vertical axis indicates the probability density, reflecting the frequency of occurrence for different error intervals.

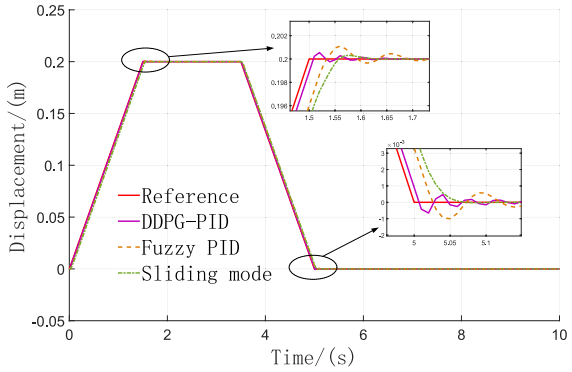


Fig. 12. Multi-stage displacement signal response.

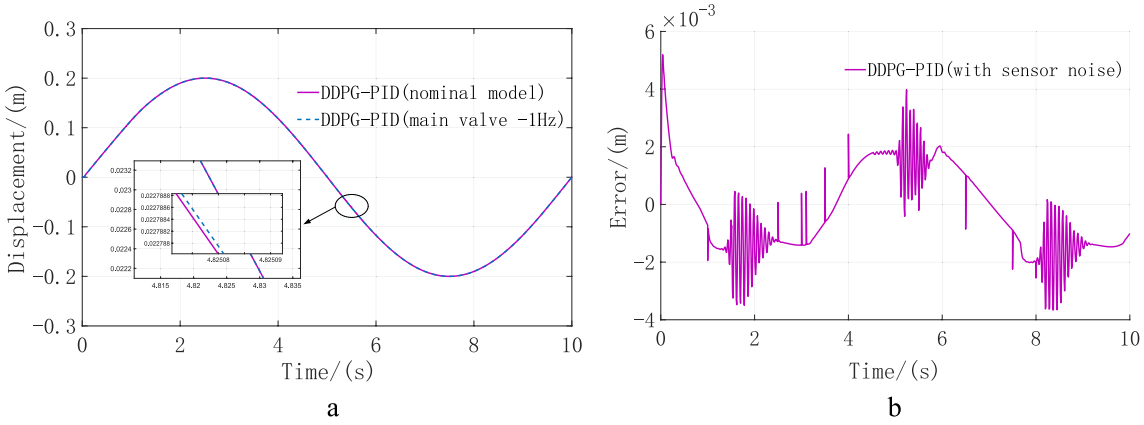


Fig. 13. Tracking performance under parameter uncertainty.

As shown in the figure, the tracking errors under the DDPG-PID controller are primarily concentrated within the ± 2 mm range, and the probability density near zero error is significantly higher compared to fuzzy PID and sliding mode control. This indicates that DDPG-PID exhibits superior tracking accuracy and response stability. In contrast, the error distributions of the fuzzy PID and sliding mode controllers are more dispersed, with the latter showing a higher probability of errors exceeding ± 4 mm.

Further statistical analysis reveals that: The proportion of errors within ± 2 mm for DDPG-PID reaches 92.6%, notably higher than 78.3% for fuzzy PID and 70.1% for sliding mode control; The maximum error of the DDPG-PID controller appears in the initial stage and is 4.7 mm, whereas the maximum errors for fuzzy PID and sliding mode control are 6.2 mm and 7.5 mm, respectively;

The error distribution of DDPG-PID is more concentrated and symmetric, indicating stronger adaptability to nonlinear disturbances.

In conclusion, the DDPG-PID control strategy outperforms traditional methods in error regulation, demonstrating the effectiveness and robustness of deep reinforcement learning in complex hydraulic system control.

Experiment 3: multi-stage displacement process simulation

To further verify the practicality of the actual project, the design of the target signal for the simulation of the injection molding step mold (0–1.5 s)—clamping (1.5–3.5 s)—mold opening (3.5–5 s) process, according to the engineering requirements of the three steps shall be completed within 5 s.

Figure 12 presents the system's response curve under a multi-stage displacement target signal, with system parameters remaining unchanged. It can be observed that the DDPG-PID controller exhibits faster convergence and smaller overshoot during both the rising and falling phases, closely following the target signal.

From Fig. 12 it follows that when the input signal is a multi-segment displacement signal, the overshoots are all within the permissible error, the sliding mode control has the largest error value at all stages, exceeding 5 mm, the fuzzy PID control has the longest regulation time, exceeding 0.3 s, and the maximum error exceeds 4 mm, while the DDPG-PID control has the smallest error and the fastest rise time, demonstrating a satisfactory control effect.

Experiment 4: simulating parameter uncertainties
Two parameter uncertainty scenarios were introduced to evaluate robustness: (1) reducing the main valve response frequency by 1 Hz, and (2) adding ± 1% Gaussian noise to the position sensor signal. Figure 13a shows that the trajectory deviation under Scenario (1) is negligible. Figure 13b reveals transient oscillations at t ≈ 3 s, 6 s, and 8.5 s due to sensor noise, but the controller quickly suppresses them, keeping the tracking error within ± 2 mm—well below the 3 mm industrial tolerance.

Figure 13a shows the displacement trajectory under Scenario (1). Compared with the standard model, the deviation is minimal and practically negligible. Figure 13b illustrates the displacement error under Scenario (2), where three distinct oscillatory intervals can be observed around t ≈ 3 s, 6 s, and 8.5 s, caused by the injected measurement noise.

The simulation results reveal that the smoothness and adjustment speed of the control signal are significantly influenced by the configuration of PID parameters. The DDPG-PID strategy learns optimal parameter combinations under dynamic conditions, effectively avoiding common issues in traditional tuning approaches, such as excessive proportional gain causing overshoot or strong integral action leading to oscillations. Thanks to the DDPG agent's sensitivity to state variations, the generated control signals maintain a good balance between fast response and signal smoothness, while also achieving better energy efficiency.

Ablation study: effects of DDPG tuning and auxiliary valve

To further evaluate the contribution of each key module in the proposed control strategy, we conducted two ablation experiments.

Experiment 1: impact of DDPG-based PID tuning
Experiment 1 focuses on assessing the impact of DDPG-based adaptive PID tuning. In this experiment, the performance of the trained DDPG-PID controller is compared with multiple fixed-gain PID controllers, all using the same control structure but differing in whether the PID parameters are updated adaptively.

Figure 14 illustrates the displacement tracking performance under a sinusoidal reference input. Although some fixed-gain PID controllers show acceptable tracking in certain intervals, the overall trajectory reveals higher deviations, especially at the beginning and end of the motion. In contrast, the DDPG-PID controller demonstrates superior adaptability and tracking accuracy, validating the benefit of reinforcement learning-based parameter tuning.

Experiment 2: impact of auxiliary valve integration
To evaluate the actual contribution of the auxiliary servo valve in a high-performance hydraulic control system, we conducted the second ablation experiment. This experiment aims to assess the impact of integrating the auxiliary valve on trajectory tracking performance, under otherwise identical conditions. The design of the auxiliary valve is inspired by dual-valve control strategies commonly used in asymmetric cylinders. Its primary role is to compensate for the initial flow imbalance between the rod and rodless chambers and to enhance the system's flow regulation capability and dynamic response, particularly at startup.

Figure 15a shows the sine wave tracking results of the controller with the auxiliary valve (auxiliary valve) and without it (only main valve). Figure 15b further presents the tracking error results from the ablation study, which demonstrate that the auxiliary valve effectively compensates for the initial flow imbalance, reducing the initial error by approximately 1 mm. Moreover, under disturbance-free conditions, it reduces the overall dynamic error of the system, resulting in a smaller error range. Detailed dynamic performance indicators are shown in Table 7.

In addition to reducing the initial error and narrowing the short-term error range, further inspection of the full trajectory reveals that the integration of the auxiliary valve contributes to a more stable dynamic response throughout the tracking process. No sustained oscillations or error drift is observed, indicating stronger long-term stability and resilience against disturbances. This feature is particularly beneficial in real-world injection molding applications, where parameter variations (e.g., oil viscosity due to temperature changes) and sensor

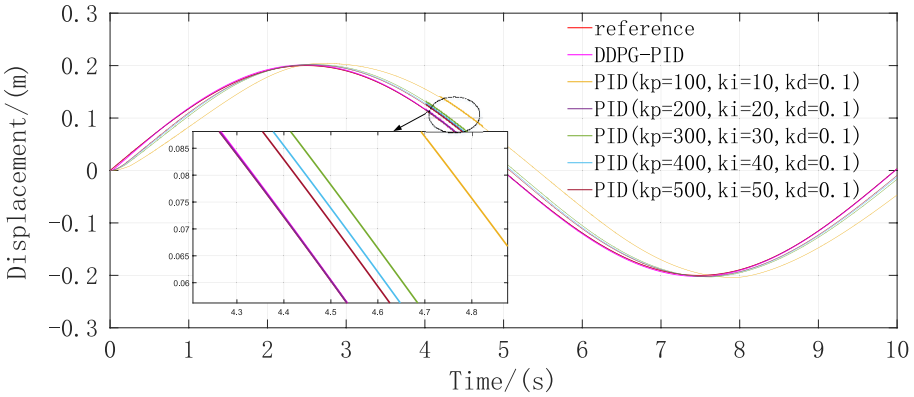


Fig. 14. Comparison of DDPG-tuned PID and fixed-gain PID controllers: ablation study under sinusoidal trajectory.

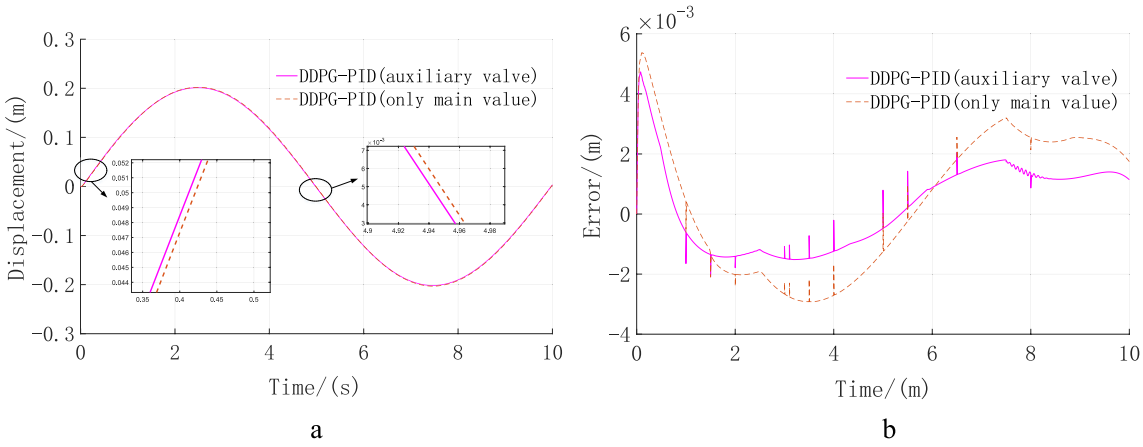


Fig. 15. Sinusoidal signal ablation experiment with and without auxiliary valve.

	Initial error (mm)	Error range (mm)
DDPG-PID (auxiliary valve)	4.37	(−1.51,1.8)
DDPG-PID (only main value)	5.36	(−2.91,3.2)

Table 7. Dynamic performance comparison of sinusoidal signal with and without auxiliary valve.

noise can compromise system consistency. By improving robustness under such conditions, the auxiliary valve enhances system reliability and reduces the need for frequent manual adjustments and maintenance.

Conclusion

This study proposes an adaptive control strategy—DDPG-PID—that integrates the Deep Deterministic Policy Gradient (DDPG) algorithm with traditional PID control to enhance position control accuracy in the clamping stage of injection molding machines. The strategy was implemented in an AMESim/Simulink co-simulation environment, where the DDPG agent continuously optimizes PID parameters through interaction with the environment, enabling adaptation to system nonlinearities and external disturbances.

Comparative experiments conducted under typical conditions, including step, sinusoidal, and multi-stage displacement inputs, validate the superior performance of the proposed method. Results show that the DDPG-PID controller outperforms fuzzy PID and sliding mode control in terms of response speed, steady-state accuracy, and disturbance suppression: In the sinusoidal tracking task, over 92.6% of the DDPG-PID errors remained within ± 2 mm, significantly better than fuzzy PID (78.3%) and sliding mode control (70.1%), as shown in Fig. 11; Under step disturbances, the DDPG-PID achieved a 74.8% reduction in peak error compared to sliding mode control, and a 27.6% improvement over fuzzy PID in terms of robustness; For multi-stage displacement inputs, the DDPG-PID exhibited smaller overshoot and faster recovery. These findings confirm that the DDPG-PID strategy possesses superior adaptability and robustness, significantly enhancing the tracking performance of hydraulic servo systems under complex operating conditions.

Future work will extend this method to multi-cylinder hydraulic systems, focusing on collaborative control among multiple actuators to address synchronization and dynamic performance challenges. By introducing Multi-Agent Reinforcement Learning (MARL), adaptive coordination among cylinders can be achieved, further improving the control accuracy and stability of hydraulic systems in nonlinear and uncertain environments. This approach offers new technical support for the intelligent control and optimization of servo-hydraulic systems.

Discussion

The proposed DDPG-PID control strategy demonstrates superior tracking accuracy and robustness under various operating conditions, primarily due to the real-time adaptive optimization of PID parameters through deep reinforcement learning. During the training process, the agent's cumulative reward showed a steadily increasing trend and eventually converged. The system consistently maintained a bounded error range when subjected to disturbances and was able to recover rapidly without noticeable deviation or oscillation, indicating strong closed-loop stability.

It is worth emphasizing that although the proposed method does not adopt a traditional model-reference adaptive control structure, it dynamically adjusts the proportional and integral gains by continuously observing the tracking error and its rate of change. This enables efficient data-driven adaptation without requiring an explicit system model. Moreover, the controller is entirely based on state-action interactions to formulate the control strategy, offering strong generalization capabilities. When applied to other hydraulic or electromechanical systems, only the observation and action spaces need to be redefined, while the core control logic can be reused directly, demonstrating good engineering scalability.

Although the simulation results confirm the effectiveness of the proposed strategy in terms of dynamic tracking and disturbance rejection, a theoretical proof of convergence and closed-loop stability is still lacking, which is a common challenge in applying deep reinforcement learning to continuous control systems. We also observed performance fluctuations of the trained policy under different initial conditions, system parameter variations, or random seeds. Although this issue was mitigated by averaging over multiple training runs, further robustness-enhancing mechanisms or transfer learning strategies are needed to ensure stability and reliability under more complex operating conditions.

Data availability

Due to the nature of this paper's research involving laboratory enterprise project content, and since the project is currently ongoing, the data and models generated or analyzed during this research period cannot be obtained from publicly available sources. If any researchers are interested, they can reasonably contact the corresponding author to obtain the information. Corresponding Author Email: lewisxin09@gmail.com, tianlei.wang@aliyun.com, 2112304125@wyu.edu.cn. Thank you for your cooperation!

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Author contributions

Z.X. and X.H. conceived the experiment(s), Z.X. and T.W. conducted the experiment(s), Z.X., W.S. and W.H. analyzed the results. G.Q. contributed to the selection of experimental components and parameter settings. W.Z. provided technical advice on experimental design and supplemental testing. G.Q. and W.Z. also provided funding. All authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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